Accuracy as a Fraction

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

AI-1 Week 23: convert fractions to percentages.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Explain why overall accuracy can hide failure on a less common class. Try the recall before opening notes.

Your learning target

I can explain why overall **accuracy** can hide failure on a less common class.

I can compare a **training-majority baseline** with class-specific results.

I can calculate an equal-class average and report its limits alongside the confusion matrix.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

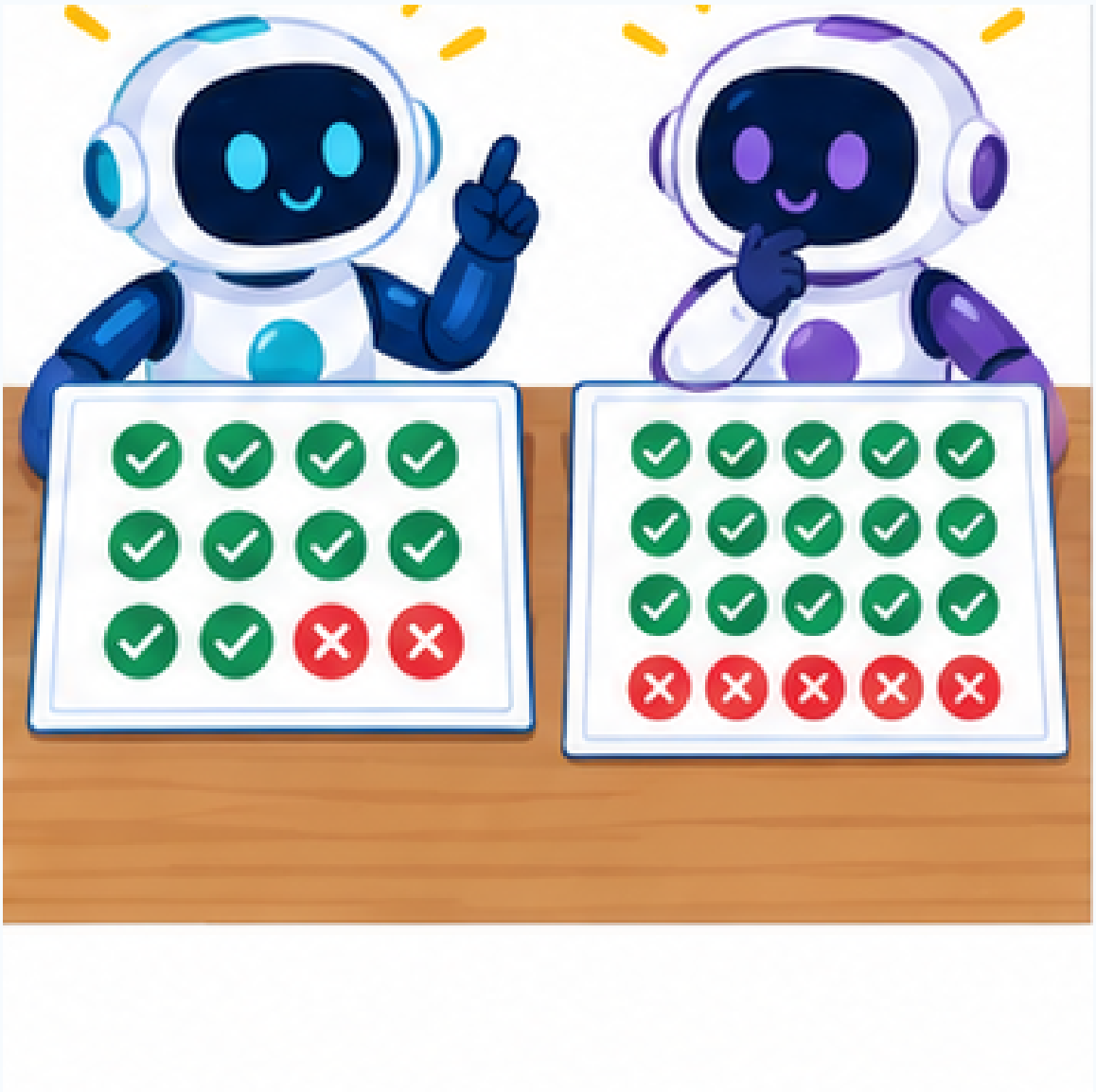
Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 12: **Accuracy** as a Fraction

Accuracy is correct **prediction**s divided by all **prediction**s.



THINKER MISSION

Calculate and compare 7/10 and 18/20 models.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

With reference rows and **prediction** columns ordered clear then blocked, where are false negatives when blocked is positive?

Readiness for Summaries with counts and proportions

If 3 of 12 responses are Yes, what fraction are Yes?

My readiness explanation

This week's end goal

By the end, I can compare **accuracy** with class-specific performance in an imbalanced evaluation.

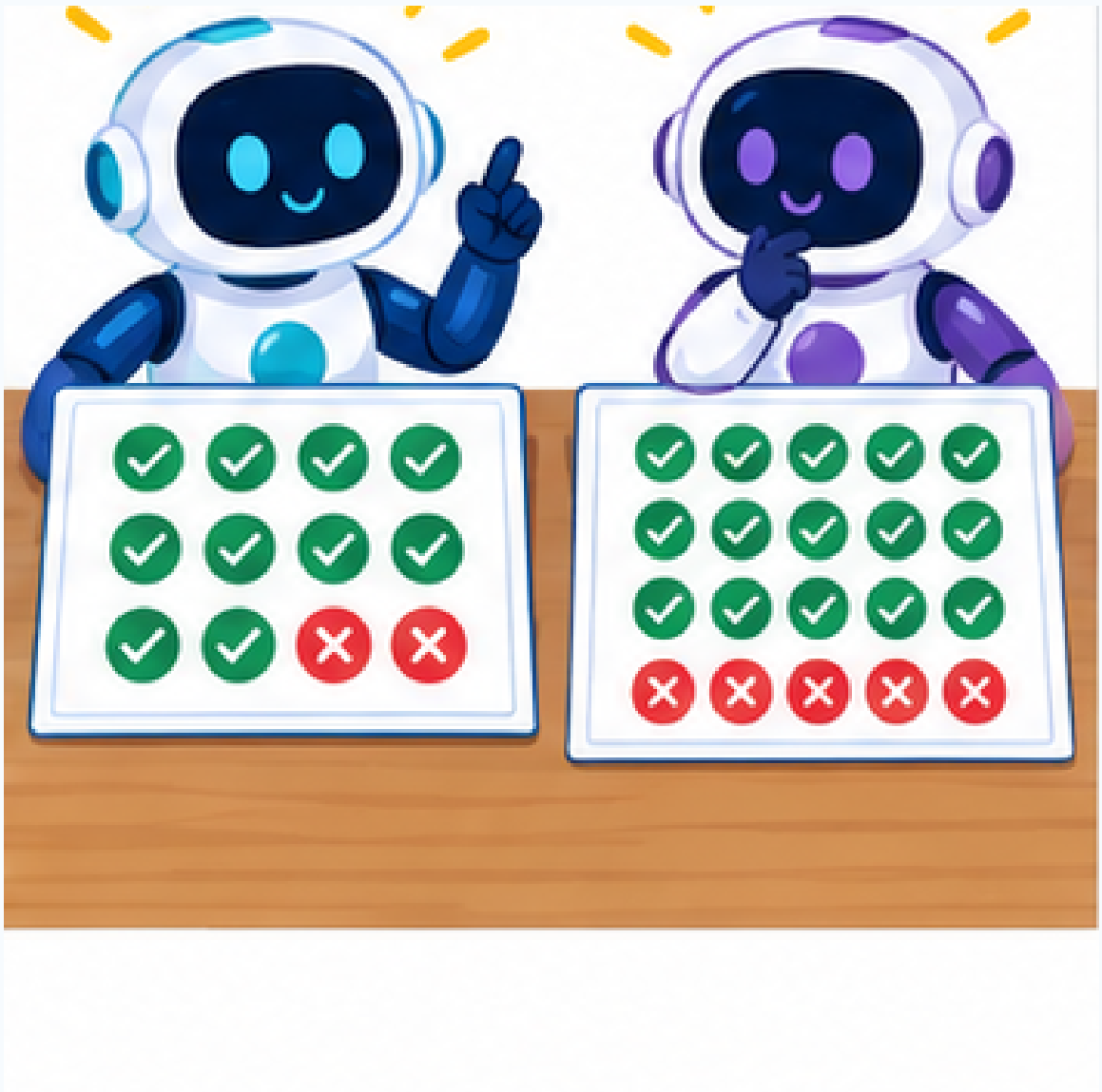
Notice the evidence

> Investigate the supplied Accuracy as a Fraction case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **accuracy** as a fraction to solve a problem. What information do they have? What decision or calculation do they make? What would **count** as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Accuracy counts all correct **prediction**s and divides by all evaluated examples. Each example has equal weight, so a common class contributes more to the score. A high result can hide failure on a rare class.

This week keeps the confusion matrix visible and checks both classes separately.

New fictional **training** data has 16 clear and 4 blocked labels. Its **majority baseline** predicts clear for every query; choose that rule from **training**, not evaluation labels.

A separate evaluation set also contains 16 clear and 4 blocked cases. A frozen candidate and the baseline are evaluated on the same 20 records.

Method and reference row	Predicted clear	Predicted blocked
Baseline reference clear	16	0
Baseline reference blocked	4	0
Candidate reference clear	12	4
Candidate reference blocked	0	4

Rows are references and columns are **prediction**s, ordered clear then blocked. Blocked is positive. Both methods get 16/20=80 percent **accuracy**. The baseline misses all four blocked cases.

The candidate finds all four but falsely flags four clear cases. The same score conceals a different tradeoff.

Apply the connected idea: Summaries with counts and proportions

A fictional poll asks whether a short activity was useful. Responses are Yes, No or Unknown. We will report Yes among known responses and also disclose unknown responses.

Choosing a **denominator** changes the question being answered.

STUDENT LESSON · 3 OF 18

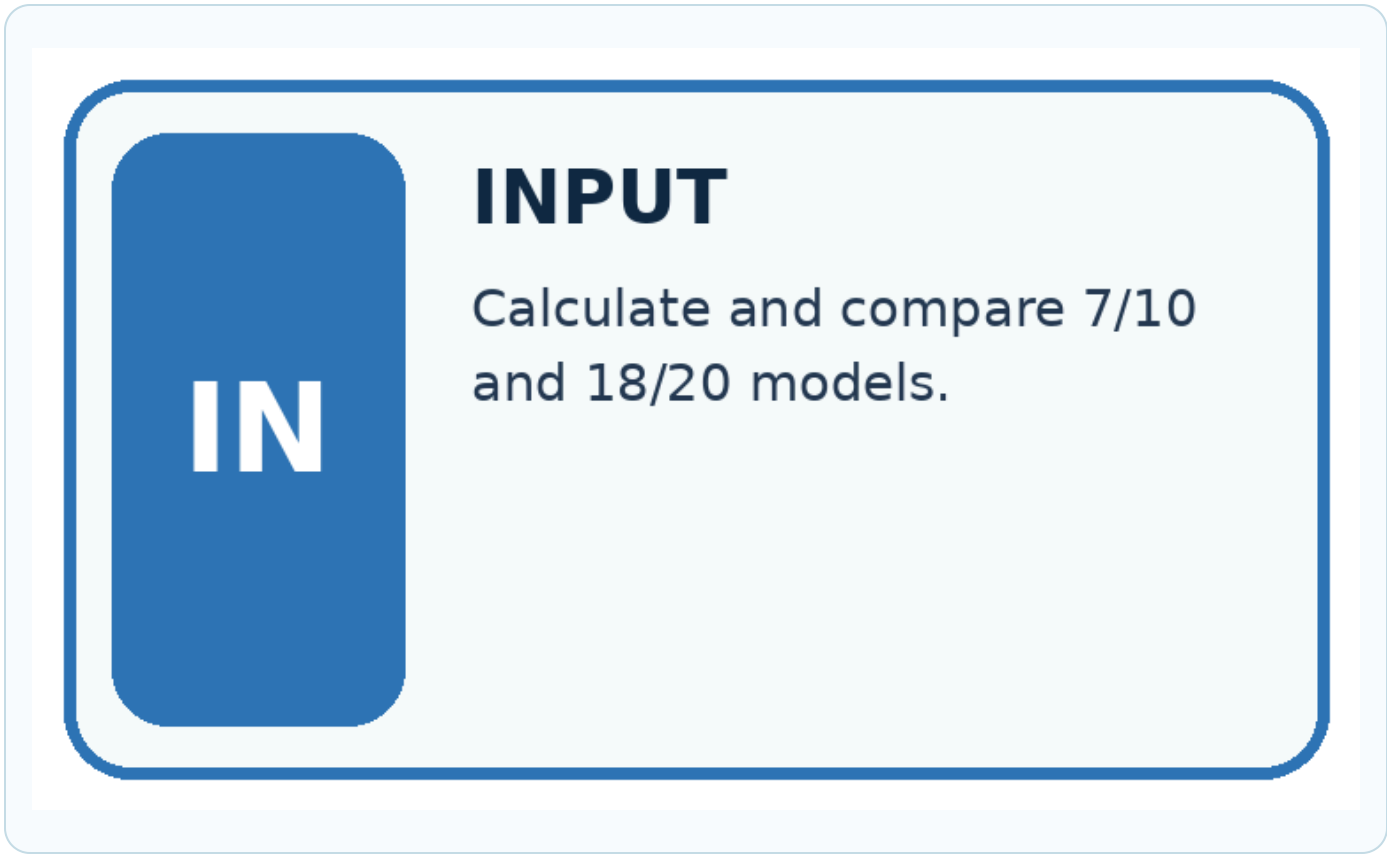
See how it works

> Follow the mechanism for Accuracy as a Fraction. Connect each labelled input, deciding step and result.

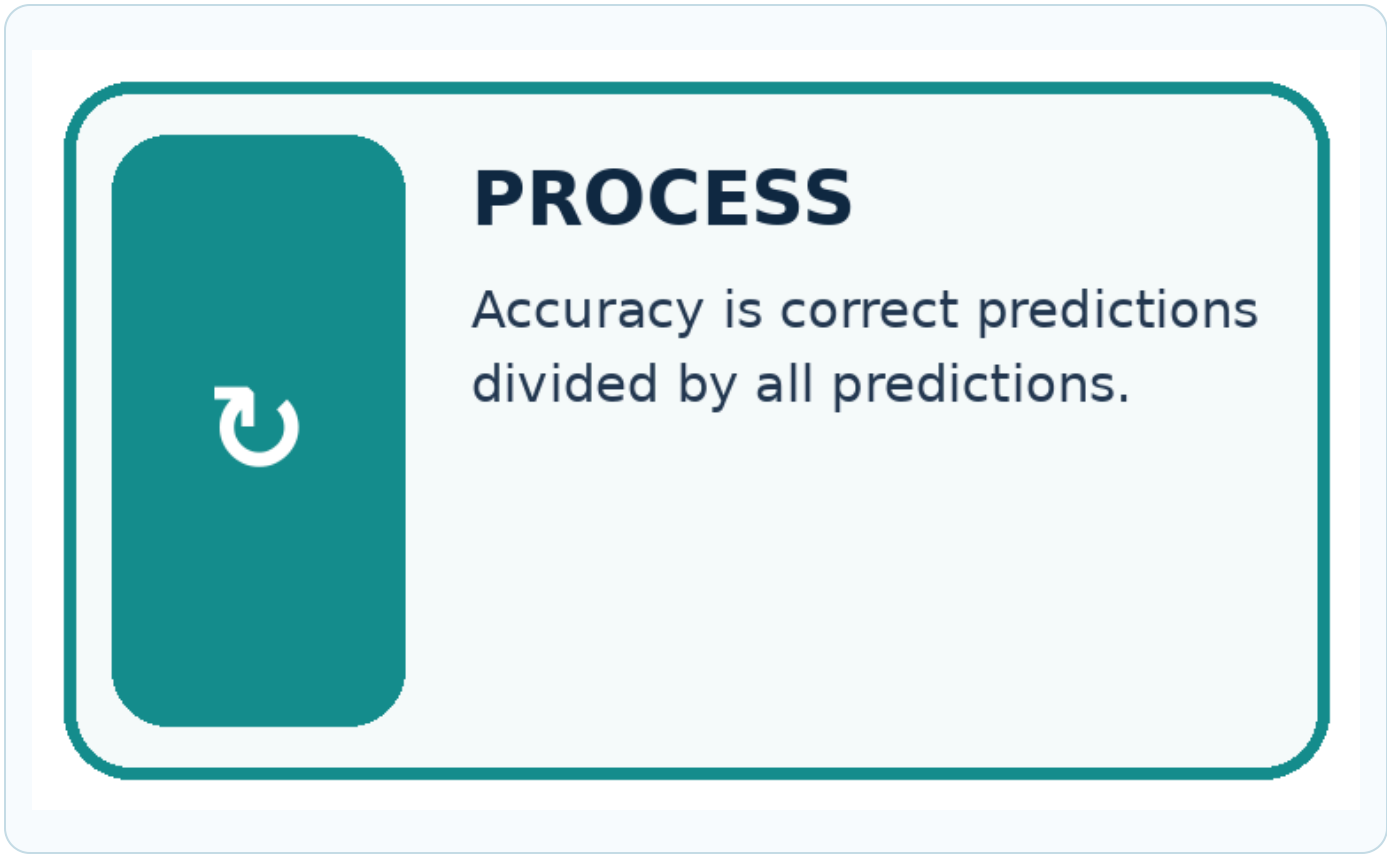
VISUAL MECHANISM

Input → Process → Output

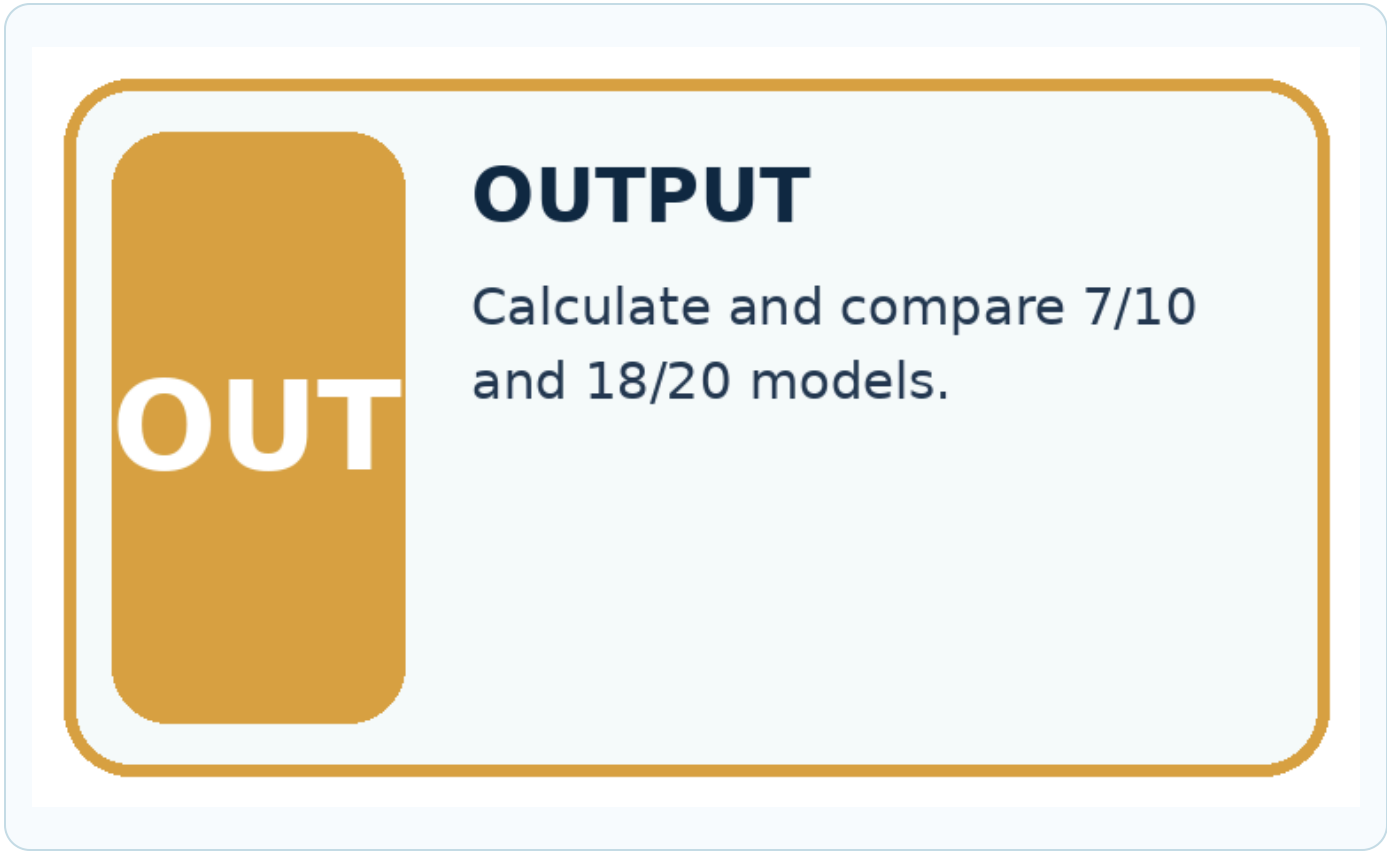
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Calculate and compare 7/10 and 18/20 models.

↓


PROCESS
Accuracy is correct predictions divided by all predictions.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
def scores(counts):
    clear_n = sum(counts[0])
    blocked_n = sum(counts[1])
    if clear_n == 0 or blocked_n == 0:
        raise ValueError("Need evaluation examples of both classes")
    correct = counts[0][0] + counts[1][1]
    accuracy = correct / (clear_n + blocked_n)
    clear_fraction = counts[0][0] / clear_n
```

```
    blocked_fraction = counts[1][1] / blocked_n
    balanced = (clear_fraction + blocked_fraction) / 2
    return accuracy, balanced

baseline = [[16, 0], [4, 0]]
candidate = [[12, 4], [0, 4]]
for name, counts in [("baseline", baseline), ("candidate", candidate)]:
    accuracy, balanced = scores(counts)
    print(name, round(100 * accuracy, 1), round(100 * balanced, 1))
```

The **output** is baseline 80.0 50.0, then candidate 80.0 87.5. In each **output** line the first **percentage** is ordinary **accuracy** and the second is **balanced accuracy**.

For the candidate, the clear-class correct fraction is 12/16=75 percent and the blocked-class fraction is 4/4=100 percent. Their equal average is 87.5 percent.

For the baseline, those fractions are 16/16=100 percent and 0/4=0 percent, averaging 50 percent. The code assumes a two-by-two table of nonnegative integer **count**s in the stated order.

It stops when either reference class is absent. Then report the observed **count**s and the unavailable class result; do not invent a **denominator** or silently substitute zero.

Apply the connected idea: Summaries with counts and proportions

```
responses = ["Yes", "No", "Yes", "Unknown", "Yes"]
yes = 0
known = 0
for response in responses:
    if response == "Yes" or response == "No":
        known = known + 1
        if response == "Yes":
            yes = yes + 1
if known > 0:
    print(yes / known)
else:
    print("No known responses")
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Yes	3	Numerator
Yes or No	4	Known denominator
All supplied	5	Includes one unknown

Read each row from left to right and connect the evidence to the stated rule.

The printed **proportion** is 0.75, or 75 percent, among four known responses. Yes among all five supplied responses is 0.6, or 60 percent. Both calculations are correct when labelled; they describe different **denominator**s.

Unknown must not silently become No.

Name the group behind a percentage

Count the responses before dividing. In the example, three say Yes, one says No and one is Unknown. There are four known responses, so Yes among known responses is 3 divided by 4, or 75 percent.

Yes among all five supplied responses is 3 divided by 5, or 60 percent. Neither number needs to be hidden. Label the question each answers, and keep the unknown **count** visible.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

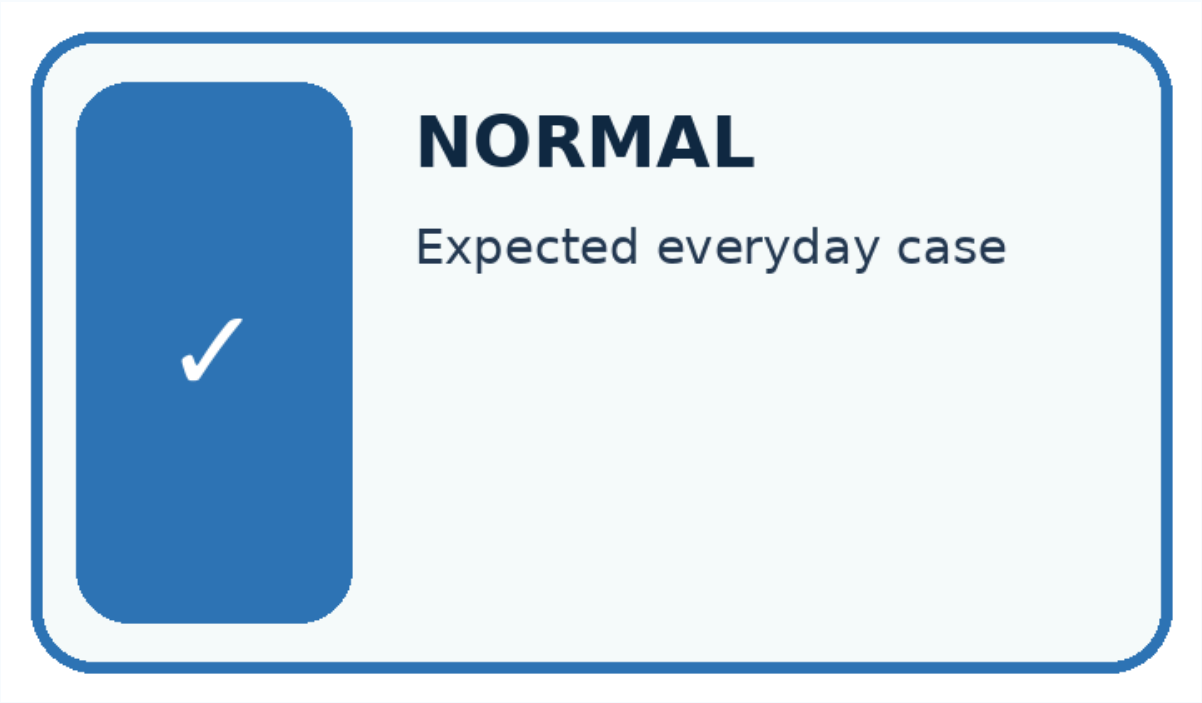
> Compare the supplied cases for Accuracy as a Fraction. Identify the detail that changes the conclusion.

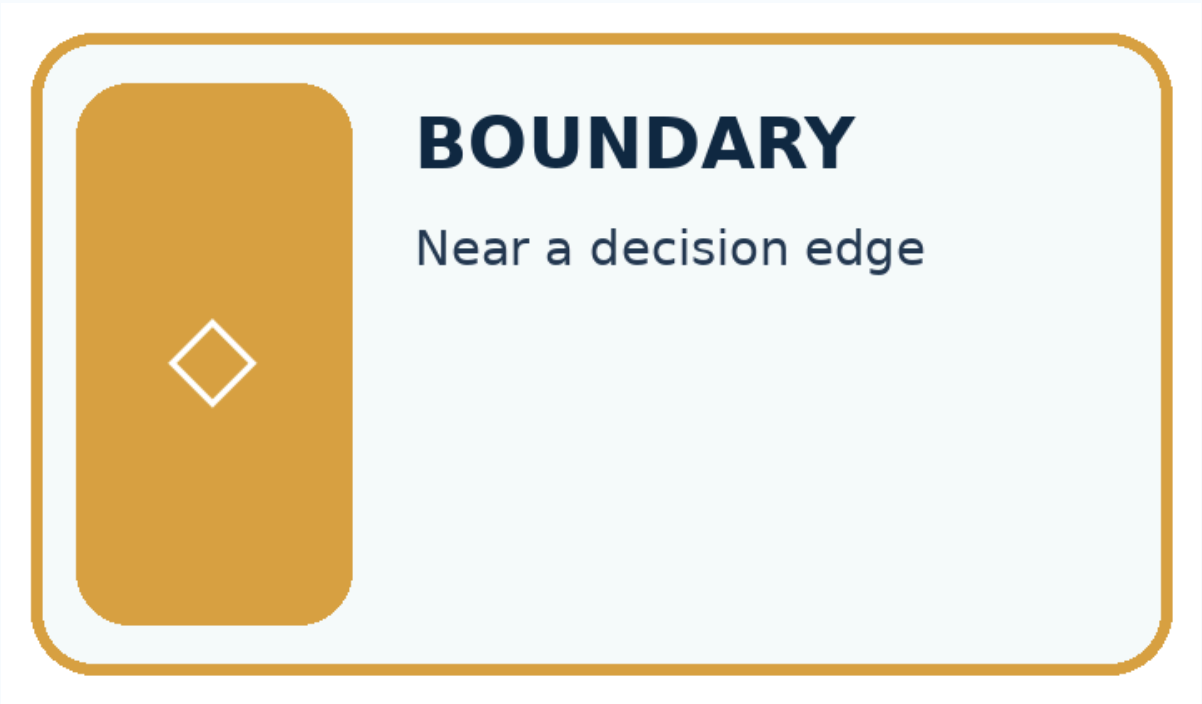
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

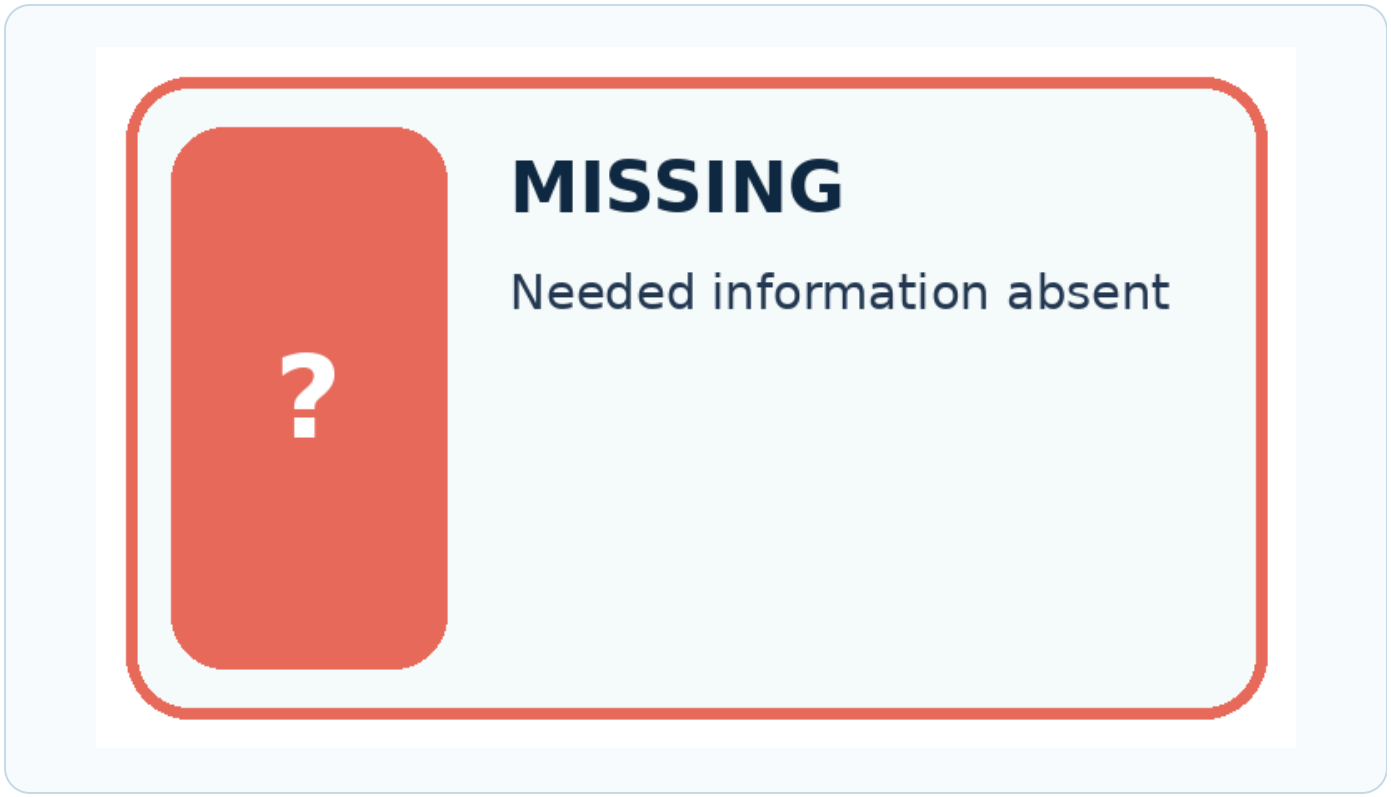
A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING
Needed information absent


→
TRANSFER
Fresh situation or user

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: A candidate instead marks every book repair needed.

Predicted result

Prediction: Recall is 4/4=100%, but **accuracy** is only 4/20=20%.

Why the result follows

Reason: All 16 not-needed books become false alarms.

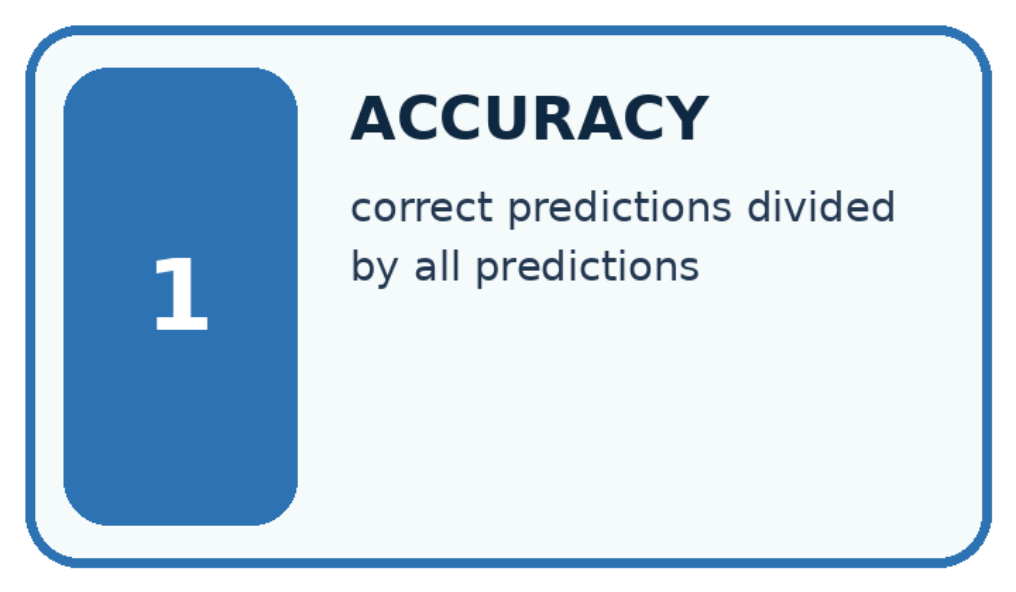
Words that unlock the idea

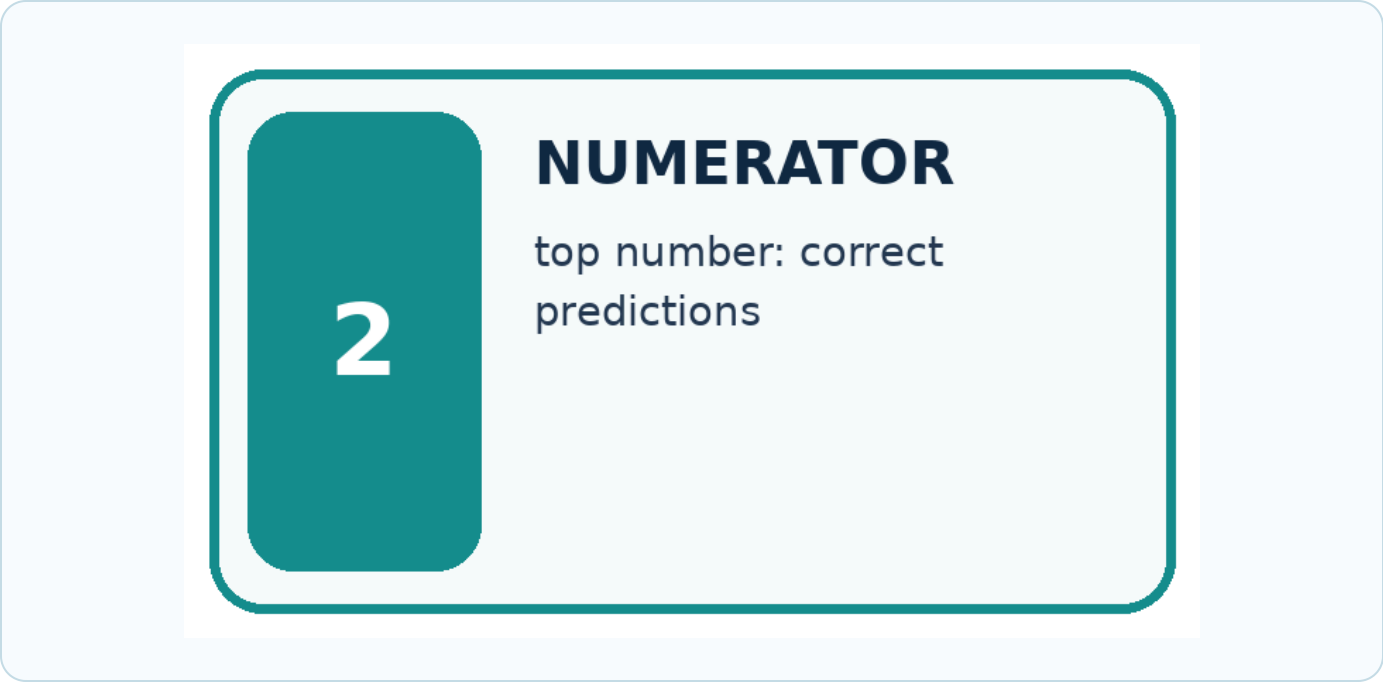
> Use the vocabulary of Accuracy as a Fraction to explain a concrete decision, rather than only memorising definitions.

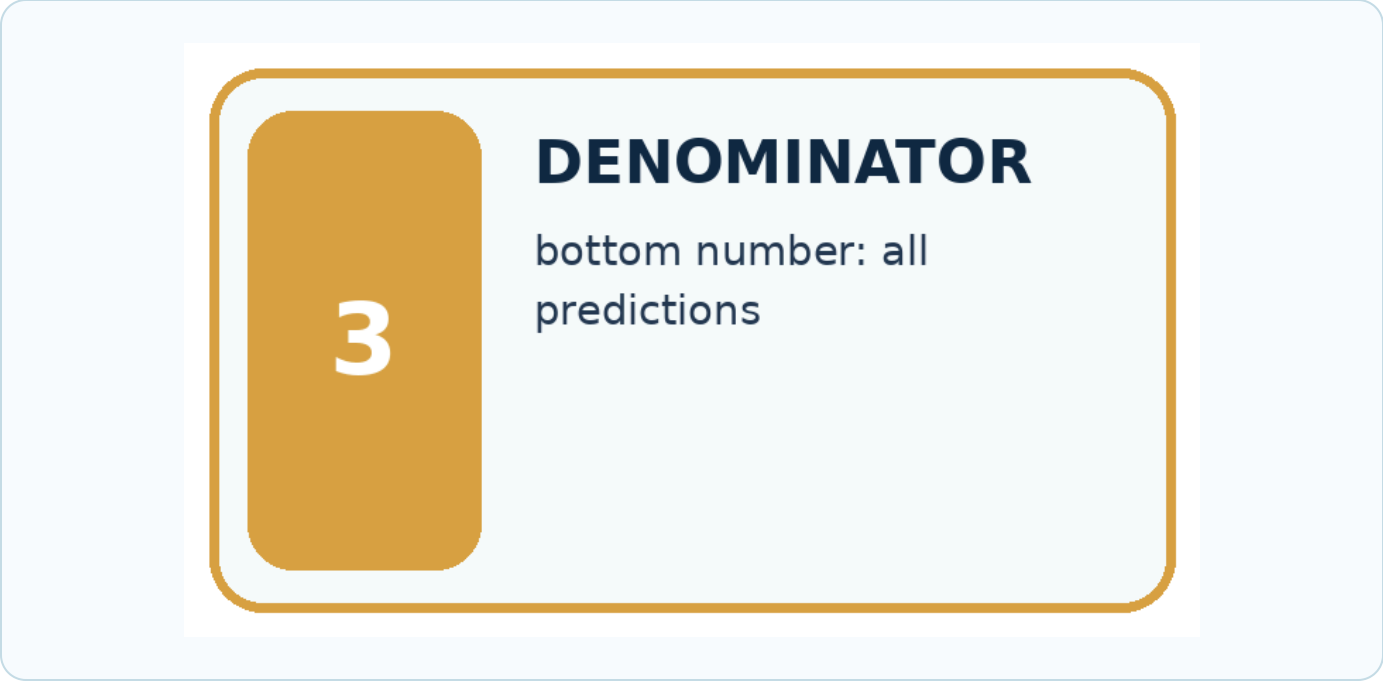
VOCABULARY LAB

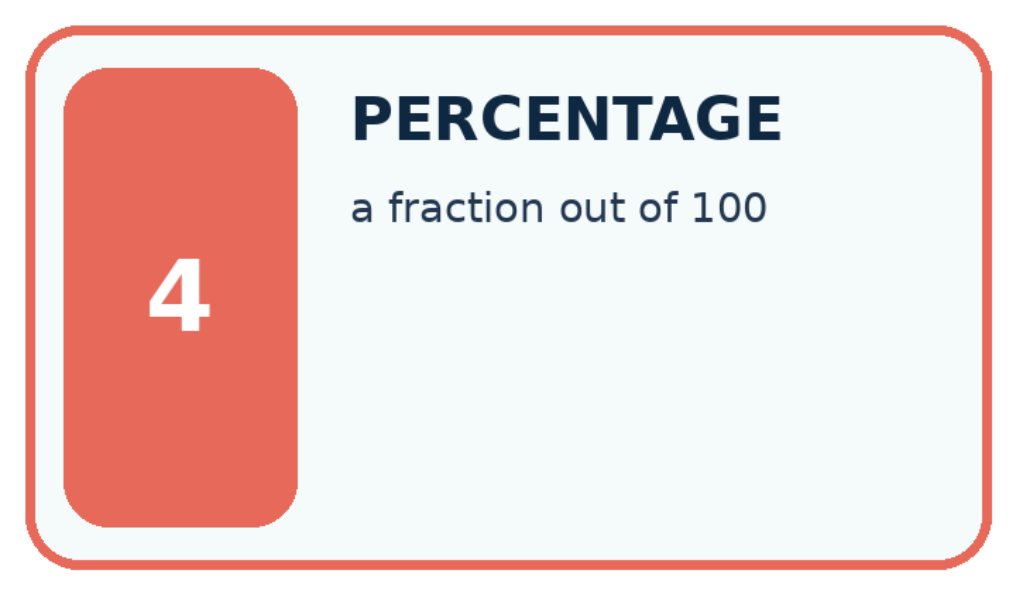
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
NUMERATOR
top number: correct predictions


3
DENOMINATOR
bottom number: all predictions



Word	My own example	Not this / boundary
accuracy	The number of correct predictions divided by the total number of scored predictions.	
numerator	The top number of a fraction, identifying the counted portion.	
denominator	The bottom number of a fraction, identifying the number of equal parts or total cases considered.	
percentage	A proportion expressed per hundred; multiply a fraction by 100 to express it as a percent.	

Words used in the closer explanation

Word	Meaning
Class imbalance	Unequal numbers of examples in different classes
Majority baseline	A rule that always predicts the most common training label
Class-specific correct fraction	Correct predictions for one reference class divided by the number of examples of that class

Word	Meaning
Balanced accuracy	The average of the class-specific correct fractions with equal weight for each class

Terms for Summaries with counts and proportions

Term	Meaning
Count	The number of matching records
Denominator	The total number used in a fraction
Proportion	A part divided by its relevant whole
Eligible	Included under a stated rule

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Accuracy as a Fraction, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 12 • WORKED EVIDENCE

1. Model A: 7 correct out of 10 → 7/10 = 70%.
2. Model B: 18 correct out of 20 → 18/20 = 90%.
3. Model B has higher measured accuracy on these tests.
4. Still ask: were the test cases fair and were some errors more costly?

1. Model A: 7 correct out of 10 → 7/10 = **70%**.

2. Model B: 18 correct out of 20 → 18/20 = 90%.

3. Model B has higher measured **accuracy** on these tests.

4. Still ask: were the test cases fair and were some errors more costly?

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Same overall score different class results

SUMMARY	ALWAYS CLEAR BASELINE	FROZEN CANDIDATE
Overall correct	16 of 20 is 80 percent	16 of 20 is 80 percent
Reference clear correct	16 of 16 is 100 percent	12 of 16 is 75 percent
Reference blocked correct	0 of 4 is 0 percent	4 of 4 is 100 percent

Overall accuracy weights classes by their number of evaluation examples. Balanced accuracy gives the two class-specific fractions equal weight.

Both methods score 80 percent. Explain which cases the baseline fails, how the candidate changes the errors, and why their balanced accuracies differ.

Worked explanation · Accuracy as a Fraction

Balanced accuracy is a metric choice, not a repair to the classifier or a guarantee of safety. It gives classes equal influence in this summary; it does not encode the actual consequences of false alarms and missed cases.

Four blocked evaluation examples are also too few for a strong general claim.

Apply the connected idea: Summaries with counts and proportions

For Yes, No, Unknown, Unknown, calculate Yes among known and Yes among all supplied.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Accuracy as a Fraction

A fictional shelf classifier predicts repair needed or not needed.

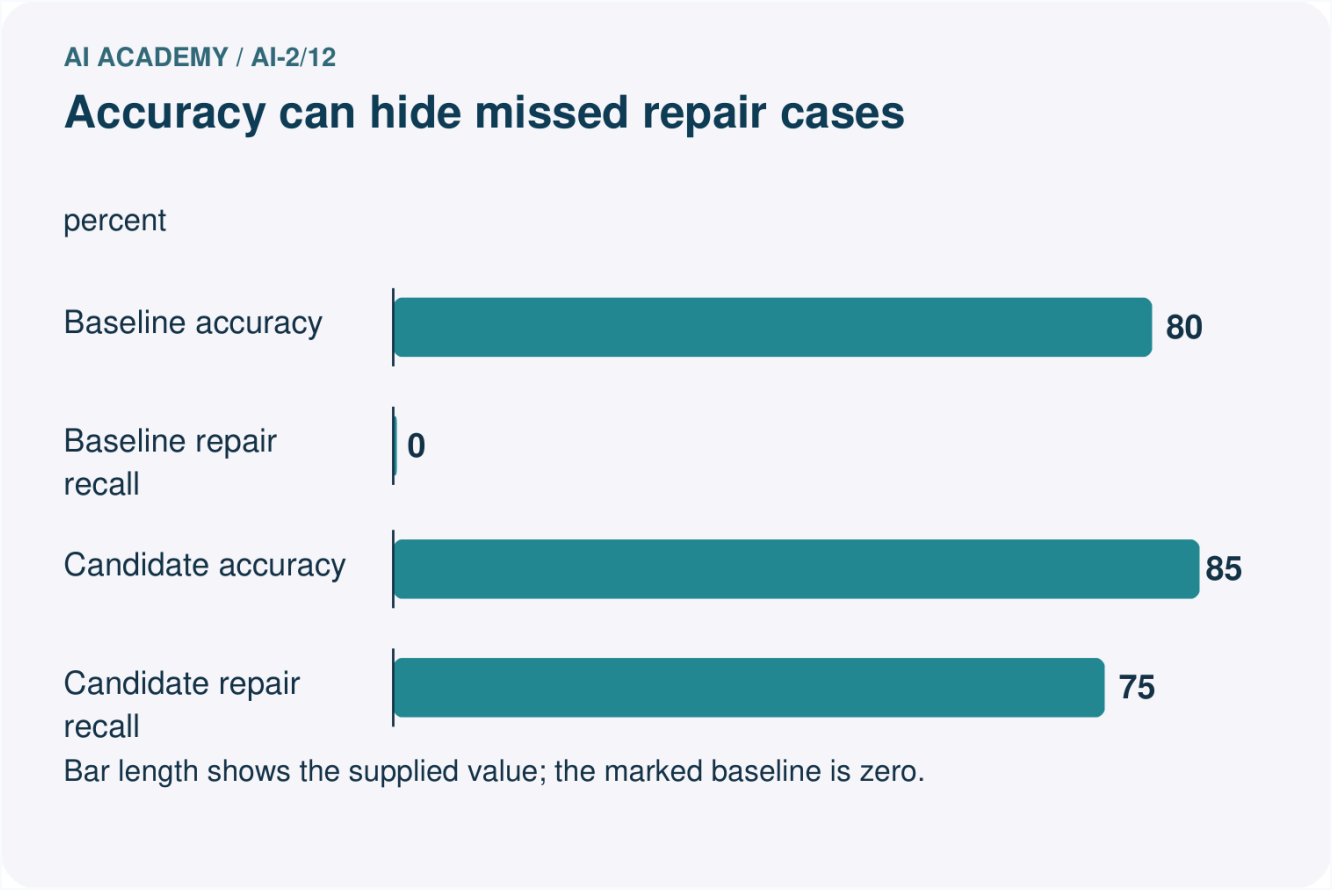
The facts we will use

Supplied facts

Evaluation reference labels: 16 not-needed and 4 needed. Positive class is needed.

Supplied facts

Always-not-needed baseline is correct on 16. Candidate has TP=3,FN=1,FP=2,TN=14.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/12

Accuracy as a Fraction

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Reference population

20 labelled cases, including four repair-needed books.

2. Baseline

16/20=80% accuracy; catches 0/4 needed cases.

3. Candidate

(3+14)/20=85% accuracy; catches 3/4=75% needed cases.

4. Compare costs

Candidate also raises two false alarms; discuss both missed repairs and extra checks.

CHANGE ONE THING

A candidate instead marks every book repair needed.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Reference population	20 labelled cases, including four repair-needed books.
2. Baseline	16/20=80% accuracy; catches 0/4 needed cases.
3. Candidate	(3+14)/20=85% accuracy; catches 3/4=75% needed cases.
4. Compare costs	Candidate also raises two false alarms; discuss both missed repairs and extra checks.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A majority-class baseline can look accurate while missing every positive case. The candidate improves overall agreement and repair detection in this table, but false alarms still matter.

Use the same cases, **denominator**s and positive-class definition for both. Do not infer that a single metric identifies the best policy for every library.

What this example does not establish

Keep the stated limit

These selected cases do not establish future rates or specify the real cost of each error.

Keep the stated limit

Read the labelled picture: Why is 100% repair detection not sufficient to declare the changed classifier best?

Keep the stated limit

Change one thing: A candidate instead marks every book repair needed. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Accuracy as a Fraction claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: The model with more correct answers always has higher accuracy, even if test sizes differ. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Compare methods on the same appropriate evaluation examples and report the **training**-derived baseline. Include ordinary **accuracy**, each class **denominator**, false positives and false negatives.

An equal-class summary adds information but cannot choose the acceptable tradeoff for the task. Keep development decisions separate from a final test.

Check the claim against the facts

A result of 4/4 correct blocked cases is 100 percent on those four examples, not proof of perfect future detection. If the evaluated set contains no blocked cases, it cannot measure blocked-case performance at all.

Next week separates precision and recall by asking which **denominator** answers each question.

A team advertises 80 percent accuracy but its classifier predicts clear on every record. What additional evidence would you request, and what can you already conclude about the four blocked cases in our worked evaluation?

Apply the connected idea: Summaries with counts and proportions

A **percentage** from a few volunteers is not automatically representative of the whole class. Report **count**s, the question, who responded and exclusions alongside the **percentage**.

Three of five known volunteers said Yes. Can you say 60 percent of all students agree?

STUDENT LESSON · 8 OF 18

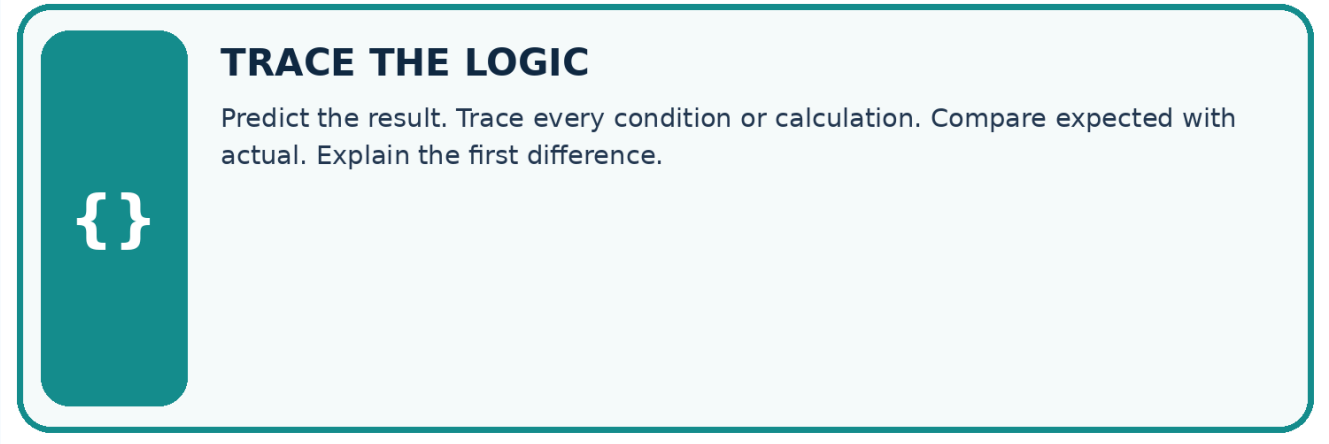
Practise together

> Practise Accuracy as a Fraction with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Consider a separate illustrative set with 8 clear and 8 blocked cases. Suppose the same within-class fractions hold: the baseline matrix is [[8,0],[8,0]] and the candidate matrix is [[6,2],[0,8]].

Ordinary **accuracy** is now 50 percent versus 87.5 percent. **Balanced accuracy** remains 50 percent versus 87.5 percent. The class mix changed the overall score even though the within-class fractions stayed fixed.

This is an arithmetic comparison, not proof that a real classifier keeps the same behavior after a population change. Report sample size, class **count**s, full matrix and how the evaluation data was chosen.

Do not remove inconvenient cases or duplicate rare evaluation cases just to improve a headline number. If **training** is resampled, keep that separate from evaluation and document it.

Apply the connected idea: Summaries with counts and proportions

Dividing by known when it equals 0 causes ZeroDivisionError. Test known > 0 before division and report no known responses if none exist. Reporting 0 percent would wrongly suggest known responses with no Yes answers.

Practise with less help: Compare both class fractions

Use workbook W2 for the connected example "**Accuracy** and **class imbalance**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Accuracy** and **class imbalance**. Proposed explanation: “High **accuracy** means both classes work well”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Summaries with counts and proportions

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

> Why is 100% repair detection not sufficient to declare the changed classifier best?

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

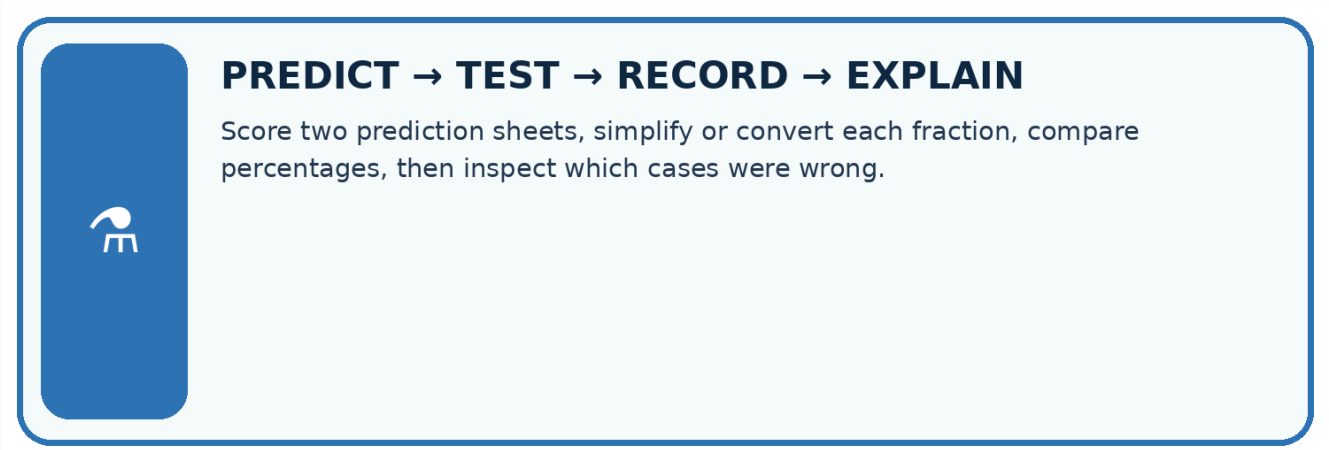
Run the investigation

> Investigate Accuracy as a Fraction. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Score two **prediction** sheets, simplify or convert each fraction, compare **percentage**s, then inspect which cases were wrong.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

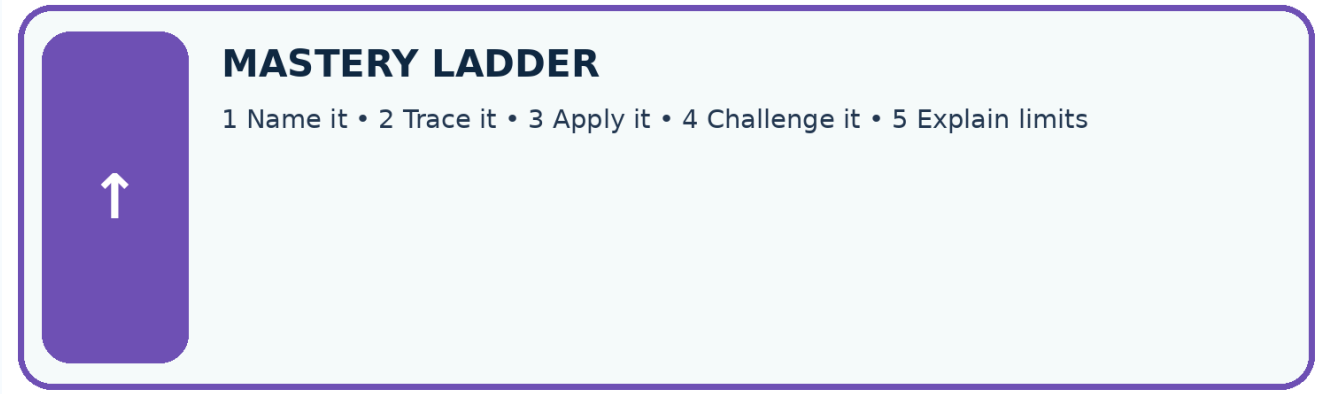
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Accuracy as a Fraction. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, **count**erexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

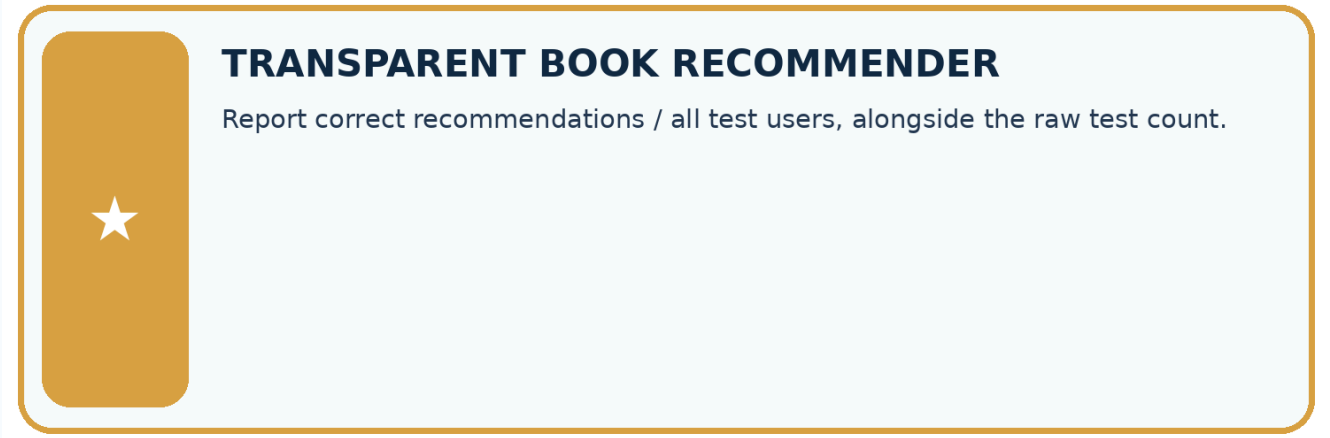
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Report correct recommendations / all test users, alongside the raw test **count**.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate **accuracy**.

STUDENT LESSON · 12 OF 18

Show what you know

Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.

✓

INDEPENDENT PROOF

Calculate and compare 7/10 and 18/20 models.

Try a more demanding case

CHALLENGE Calculate and compare 7/10 and 18/20 models.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Accuracy as a Fraction case for supported practice. Repeated examples do not become fresh assessments.

Week 12 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Nine of twelve held-out classifications are correct.

The question

Compute **accuracy** and state the **denominator**.

Worked explanation

9/12=3/4=75%. **Count** all scored cases, including errors.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

All scored cases belong in the denominator

Evidence or stage	Value or result
Correct classifications	9/12

Nine of twelve held-out classifications are correct.

Compute **accuracy** and state the **denominator**.

9/12=3/4=75%. **Count** all scored cases, including errors.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Assess a new comparison independently

New **training** labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records.

The always-clear baseline is [[18,0],[2,0]] and candidate is [[15,3],[1,1]], with reference rows and clear then blocked on both axes. Calculate ordinary **accuracy**, both class fractions and **balanced accuracy** for each.

Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 12 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A classifier gets 9 of 12 cases right. Another gets 8 of 10 right. Calculate both accuracies and identify the larger fraction.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Report correct recommendations / all test users, alongside the raw test **count**.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> Class A gets 8/10 cases right and class B gets 18/30 right. Compute each rate and the combined rate. Why is averaging the two **percentage**s misleading here?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Accuracy as a Fraction, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional shelf classifier predicts repair needed or not needed.

By the end, I can	How I will show it
Explain why overall accuracy can hide failure on a less common class	Explain the deciding step, show a trace or test, and state one limit.
Compare a training-majority baseline with class-specific results	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Calculate an equal-class average and report its limits alongside the confusion matrix	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Accuracy as a Fraction without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain why overall **accuracy** can hide failure on a less common class

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional shelf classifier predicts repair needed or not needed.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Candidate also raises two false alarms; discuss both missed repairs and extra checks.

1. Reference population

2. Compare costs

3. Candidate

4. Baseline

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare costs Candidate also raises two false alarms; discuss both missed repairs and extra checks.

A majority-class baseline can look accurate while missing every positive case. The candidate improves overall agreement and repair detection in this table, but false alarms still matter.

Use the same cases, **denominator**s and positive-class definition for both. Do not infer that a single metric identifies the best policy for every library.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A candidate instead marks every book repair needed. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Recall is 4/4=100%, but **accuracy** is only 4/20=20%.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Recall is 4/4=100%, but **accuracy** is only 4/20=20%. All 16 not-needed books become false alarms.

Repair a missing building block

Review **Training** and Testing (ai-2/11). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan **count**s?.

Return to this week and explain what you changed.

Review Features and Distractors (ai-2/9). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Does this one pair prove the classifier always ignores subject?

Explain a narrower claim.. Return to this week and explain what you changed.

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

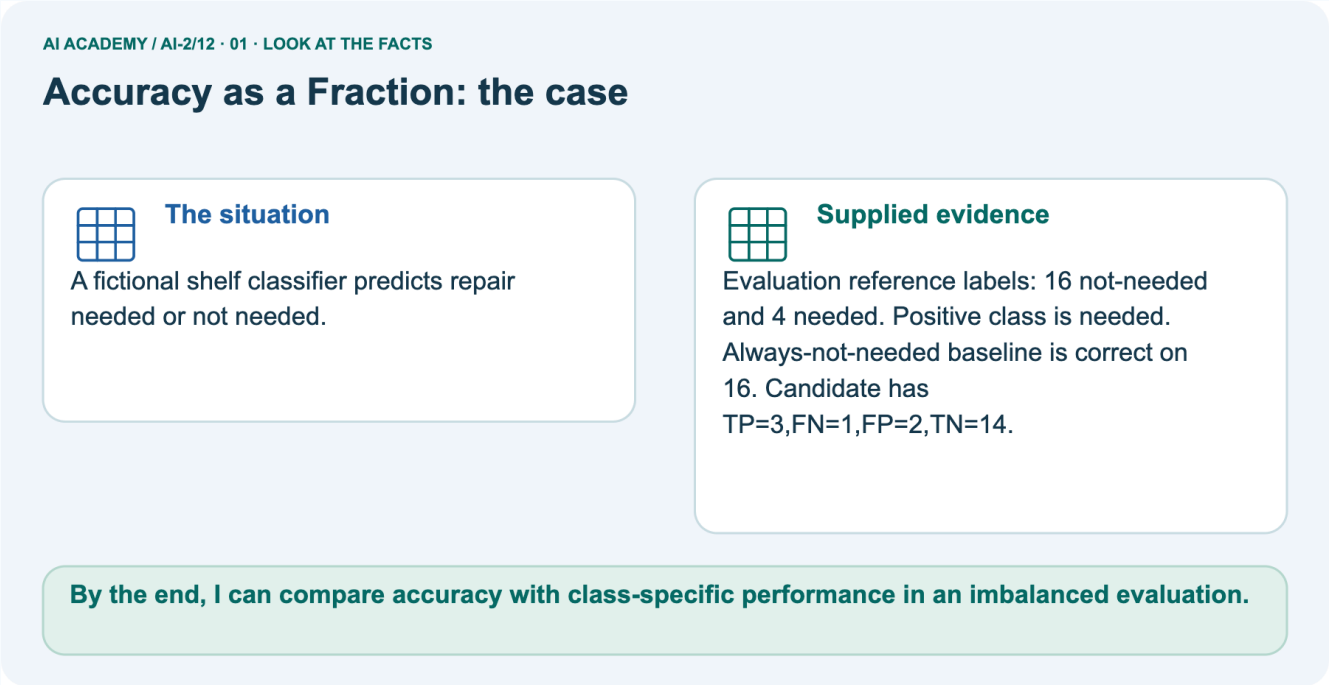
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Accuracy as a Fraction graphic using its labels, deciding evidence and relevant limitation.

01. Accuracy as a Fraction: the case



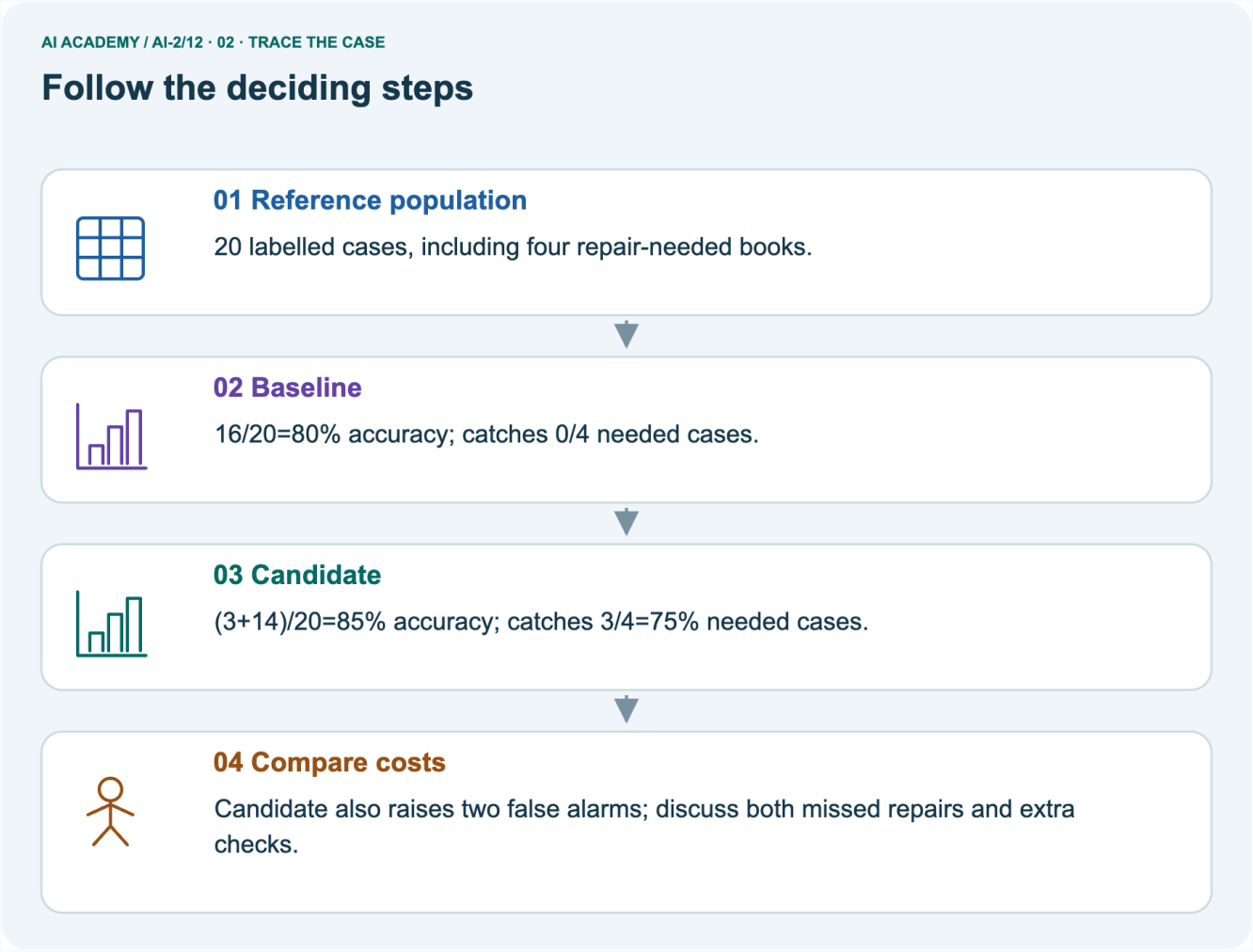
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional shelf classifier predicts repair needed or not needed. Supplied facts: Evaluation reference labels: 16 not-needed and 4 needed. Positive class is needed. Always-not-needed baseline is correct on 16.

Candidate has TP=3,FN=1,FP=2,TN=14.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

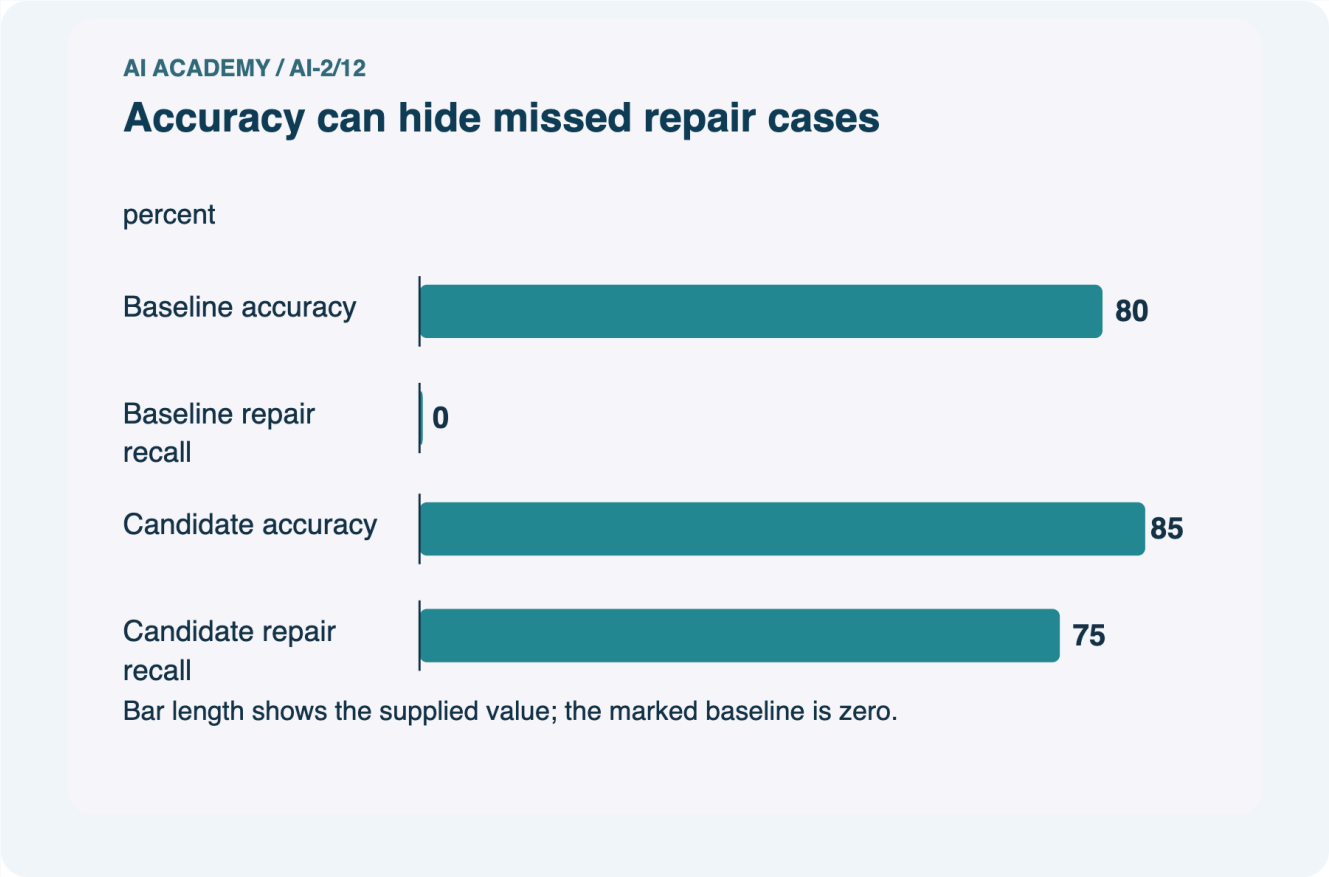


Follow the labelled picture with the matching explanation. Open the vector diagram

Reference population: 20 labelled cases, including four repair-needed books. → Baseline: 16/20=**80% accuracy**; catches 0/4 needed cases. → Candidate: (3+14)/20=85% **accuracy**; catches 3/4=75% needed cases. → Compare costs: Candidate also raises two false alarms; discuss both missed repairs and extra checks.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Accuracy can hide missed repair cases

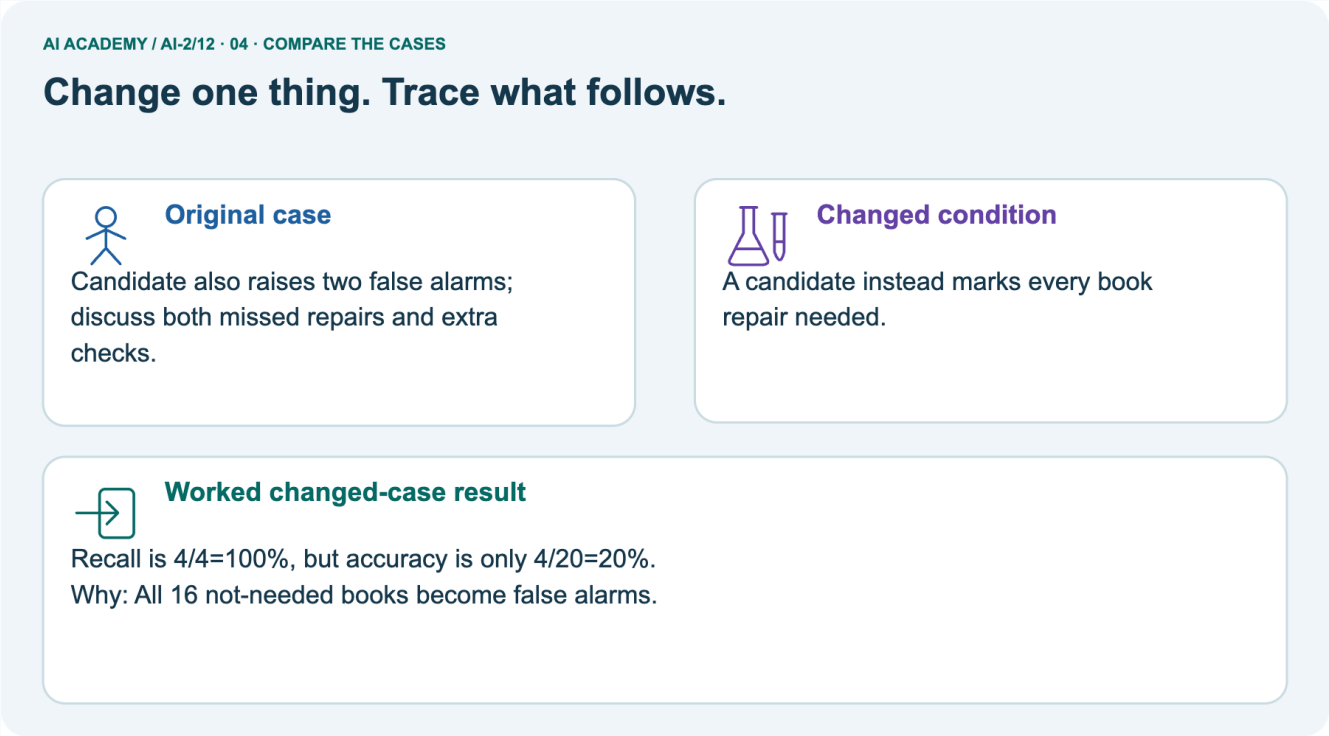


Follow the labelled picture with the matching explanation. Open the vector diagram

Accuracy can hide missed repair cases. Units: percent. Baseline **accuracy**: 80. Baseline repair recall: 0. Candidate **accuracy**: 85. Candidate repair recall: 75.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



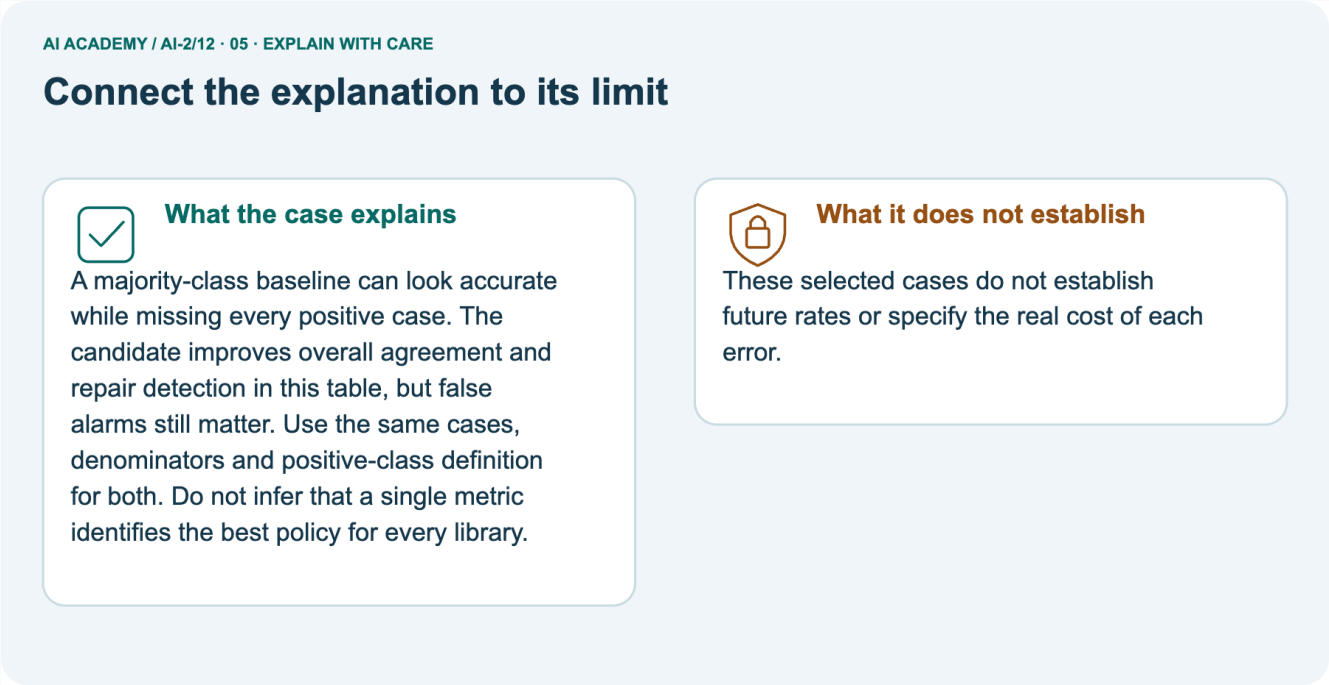
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Candidate also raises two false alarms; discuss both missed repairs and extra checks. Changed condition: A candidate instead marks every book repair needed.

Expected result: Recall is 4/4=100%, but **accuracy** is only 4/20=20%. Reason: All 16 not-needed books become false alarms.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

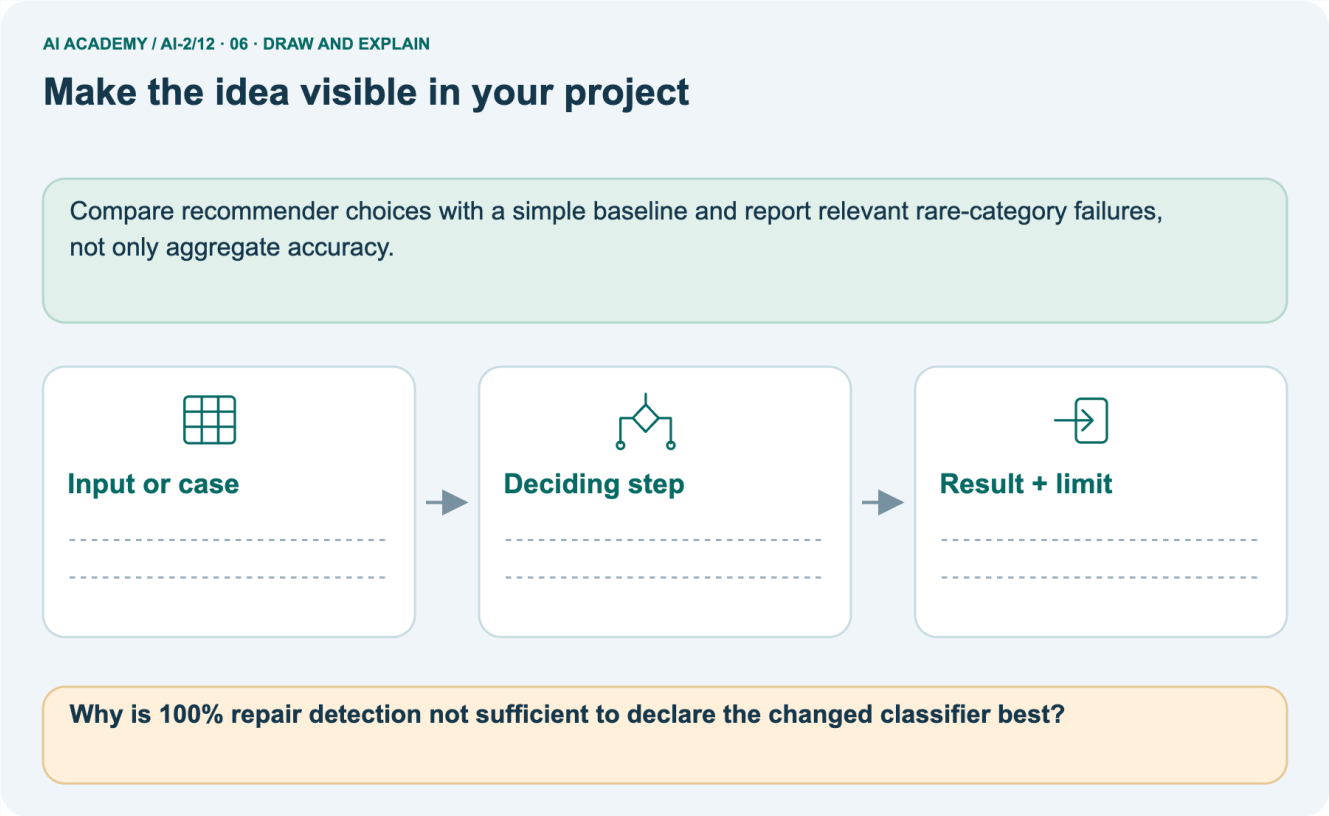
A majority-class baseline can look accurate while missing every positive case. The candidate improves overall agreement and repair detection in this table, but false alarms still matter.

Use the same cases, **denominator**s and positive-class definition for both. Do not infer that a single metric identifies the best policy for every library.

Limit: These selected cases do not establish future rates or specify the real cost of each error.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate **accuracy**. Draw your **input**, deciding step, result and limit.

Supported question: Why is 100% repair detection not sufficient to declare the changed classifier best?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.